

Positive Square Energy of Every Octacyclic Cactus

26 July 2026

Abstract

Let G be a connected octacyclic cactus on n vertices. We prove $s^+(G) > n$. Sharp block-additive doubly nonnegative optimization reduces the problem to T^7Q and T^6PP , where $T = C_3$, $P = C_5$, and Q is an arbitrary cycle. For disconnected shared-cut graphs, exact colored-partition censuses give $44 = 42 + 2$ and $76 = 70 + 6$ proper rows. The structural T^7Q endpoint is closed by a standalone packing-one Sachs lemma whose hypothesis is checked on a common-cut triangle bouquet. The entry-locked $T^6P \mid P$ endpoint is closed by the corrected strict-last-bridge certificate $877 = 861 + 16$, including all sixteen marked profiles and the distinction between a shared-cut TTP packet and a common-cut bouquet. For one shared-cut cluster, exact incidence censuses leave only the common-cut T^7Q bouquet and six of the 2116 types in T^6PP ; the common-cut Schur–Sachs theorem and six explicit router resolutions close them. Every partition is induced and assigns every connector remnant, router interval, shared cut, and attached tree exactly once. The proof expressly does not use the retracted all-rank rooted hostile-cycle guard or the superseded uncut $877 = 868 + 9$ certificate.

1 Definitions and validity boundary

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles. Its cyclomatic number equals its number of cyclic blocks. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For every vertex partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_r$, induced-subgraph superadditivity gives

$$s^+(G) \geq \sum_i s^+(G[V_i]), \quad \sigma(G) \geq \sum_i \sigma(G[V_i]). \quad (1)$$

This is the theorem of Akbari, Kumar, Mohar, and Pragada [1]. Akbari, Kumar, Mohar, Pragada, and Zhang conjectured that $s^+(G) \geq |V(G)|$ for every connected graph with at least $|V(G)| + 1$ edges [2, Conjecture 1.2]. Since an octacyclic graph has $|E(G)| = |V(G)| + 7$, Theorem 7.1 proves their conjecture, strictly, for every connected octacyclic cactus.

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{4}, \quad \delta_q = \sec(\pi/q) - 1 < 1.$$

The symbol Q denotes C_q . It is called hostile only when $q \equiv 1 \pmod{4}$; otherwise its unicyclic packet is nonnegative.

This manuscript uses only the corrected provenance summarized in [12]: the strict-last-bridge $877 = 861 + 16$ certificate [14], the standalone packing-one hostile-cycle lemma [13], the exact fully shared census and six replacements [15, 16], and the scalar common-cut theorem [17]. We explicitly reject and do not use the retracted all-rank rooted hostile-cycle guard. In particular, we never assert that a maximum triangle packing makes every nearest-cycle Voronoi territory packing one. We also do not use the superseded uncut $877 = 868 + 9$ (E1–E9) computation; its connector ownership was defective. Every occurrence of “packing one” below is the standalone lemma with its hypothesis verified directly.

There are two intentionally different terms. A *shared-cut TTP* packet is a connected packet in which its two triangles share a cut; its pentagon belongs to the packet but need not contain that cut. A *common-cut T^kP* or *common-cut T^kPP* packet has one pivot contained in every named cycle. The former is never silently treated as the latter.

2 Packet bounds and territory operations

We use the established lower-rank and packet estimates

$$\sigma(H) \geq 0 \quad \text{if } H \text{ is bicyclic or tricyclic,} \quad (2)$$

$$\sigma(H) > 0 \quad \text{if } H \text{ has rank 4, 5, 6, or 7,} \quad (3)$$

$$\sigma(P) \geq -\delta, \quad \sigma(Q) \geq -\delta_q \quad \text{for hostile } Q, \quad (4)$$

$$\sigma(TP) > 1 - \delta, \quad (5)$$

$$\sigma(PP) > 0 \quad \text{for one shared-cut cluster,} \quad (6)$$

$$\sigma(TPP) > \frac{3}{2} \quad \text{for one shared-cut cluster,} \quad (7)$$

$$\sigma(TTP) > 2 - \delta \quad \text{when the two triangles share a cut.} \quad (8)$$

All permit arbitrary finite trees at arbitrary core vertices; TP permits an arbitrary bridge connector. The hostile unicyclic bound is [6, Lemma 1.1(2)], the mixed TP bound is [5, Theorem 3.1], and the complete bicyclic and tricyclic bounds are [5, Theorem 5.1] and [6, Theorem 5.1]. The shared-cut PP bound is [5, Theorem 4.1]; the shared-cluster TPP and shared-cut TTP bounds are [6, Proposition 4.2] and [6, Proposition 3.1]. The strict rank-four input is the repaired tetracyclic theorem [7, Theorem 5.1]: its proof uses the exact twenty-core C_{3355} audit and common-root phase repair, not the false raw C_{3335} coefficient certificate. The strict rank-five, rank-six, and rank-seven inputs are [8, Theorem 8.1], [9, Theorem 9.1], and [10, Theorem 9.1]. If A_r is one connected shared-cut cluster of r triangles, successive incidence-leaf openings from the corrected rank-four matching injection give

$$\sigma(A_r) > b_r, \quad (b_1, \dots, b_7) = (0, 1, 2, 3, 2, 1, 0). \quad (9)$$

In particular, $\sigma(A_4) > 3$ is not obtained from a multiplicity-blind Sturm count. The complete recurrence, including arbitrary attached trees, is [10, Lemma 3.1]; its rank-four exceptional base is the exact matching-injection result [11, Proposition 4.11].

Two analytic results are essential.

Lemma 2.1 (Standalone packing-one hostile packet). *Let a connected cactus have one hostile $Q = C_q$ and $a \geq 1$ triangles, no two of which are vertex-disjoint. The Q may meet the triangular part at a root or be joined to its root by an arbitrary positive path. Arbitrary trees may occur everywhere, including on that path. Then*

$$s^+(H) - s^-(H) > -2\delta_q, \quad \sigma(H) > a - \delta_q.$$

Proof. Let $S = V(Q)$ and let the spine be the cyclic blocks and their minimal joining paths. Eliminate every component off the spine by the exact signless matching message

$$M_{u \rightarrow v}(t) = \frac{Z_{T_{u \rightarrow v}}(t)}{Z_{T_{u \rightarrow v} - u}(t)} = t + \sum_w M_{w \rightarrow u}(t)^{-1} \geq t.$$

This replaces each spine activity by $\alpha_v = t + y_v \geq t$ and removes a common positive factor. Write $Z_F(\alpha)$ for the resulting signless matching partition. The grouped Sachs expansion on the imaginary axis is

$$\Psi_H(t) = R + 2i(B - A),$$

where, after normalization,

$$B = Z_{H-S}(\alpha), \quad A = \sum_T Z_{H-V(T)}(\alpha), \quad R = Z_H(\alpha) + 4 \sum_{T \cap Q = \emptyset} Z_{H-(S \cup V(T))}(\alpha) > 0.$$

Packing one is used exactly here: no Sachs collection contains two triangles; the last sum is precisely the admissible Q -triangle terms. If $Z_q(t)$ is the bare matching partition of Q , split the matchings in $Z_H(\alpha)$ by whether they use a boundary edge of S . Then

$$Z_H(\alpha) = Z_Q(\alpha|_S)B + E, \quad E \geq 0, \quad Z_Q(\alpha|_S) = Z_q(t) + L, \quad L \geq 0,$$

and hence

$$R - Z_q(t)(B - A) = E + LB + Z_q(t)A + 4 \sum_{T \cap Q = \emptyset} Z_{H-(S \cup V(T))}(\alpha) > 0.$$

If $B - A \leq 0$ the argument is nonpositive; otherwise division by $RZ_q(t) > 0$ gives the same conclusion. Thus the continuous argument of $\Psi_H(t)$ is strictly smaller than $\arctan(2/Z_q(t))$, the argument for isolated Q . The signed Coulson identity

$$s^+(F) - s^-(F) = -\frac{4}{\pi} \int_0^\infty t \operatorname{Arg} \Psi_F(t) dt$$

and $s^+(C_q) - s^-(C_q) = -2\delta_q$ give the first inequality. Since $|E(H)| = |V(H)| + a$ and $s^+ + s^- = 2|E|$, the second follows. This is the standalone grouped-Sachs proof of [13]; it has no all-rank or Voronoi premise. $\square$

Lemma 2.2 (Common-cut Schur–Sachs theorem). *For a one-vertex bouquet, uniformly over arbitrary attached trees,*

$$\sigma(T^k Q) > k - \delta_q \quad (q \equiv 1 \pmod{4}), \quad (10)$$

$$\sigma(T^k P) > k - \delta \quad (k \geq 1), \quad (11)$$

$$\sigma(T^k Q) > k \quad (q \text{ even or } q \equiv 3 \pmod{4}), \quad (12)$$

$$\sigma(T^k PP) > k + 1 - \frac{4}{3\sqrt{13}}. \quad (13)$$

Proof. After exact Schur elimination of attached trees, each core activity is $a_v(t) = t + y_v(t)$ with $y_v \geq 0$. If A_j is the matching partition of the private path in lobe j and B_j is the contribution obtained by matching the pivot into that lobe, the normalized determinant is the scalar expression

$$W(t) = a_x + \sum_j \frac{B_j + \omega_{\ell_j}}{A_j}, \quad \omega_\ell = -2i^{-\ell}.$$

For $T^k Q$, put

$$X = a_x + B_Q/A_Q + \sum_{j=1}^k B_j/A_j > 0.$$

For hostile Q the imaginary part is $2(1/A_Q - \sum_j 1/A_j)$, while $A_Q X > Z_q(t)$ because $k \geq 1$, $B_j > 0$, and all activities are at least t . If that imaginary part is nonpositive the phase is nonpositive; otherwise

$$\frac{2(1/A_Q - \sum_j 1/A_j)}{X} < \frac{2}{A_Q X} < \frac{2}{Z_q(t)}.$$

Thus its argument is strictly below the isolated hostile- Q argument, and is negative for the other congruence classes. Coulson integration gives (10) and (12). Taking $Q = P = C_5$ gives the explicit specialization (11); this is the common-cut $T^k P$ estimate used in L1–L12 and U2–U3. For $T^k PP$, write $X_{PP} = a_x + B_1/A_1 + B_2/A_2$ and $Y_{PP} = 2/A_1 + 2/A_2$. Adding triangles strictly increases the positive real part and decreases the imaginary part, so its phase is strictly below that of the two-pentagon core. The exact positive-coefficient inequality

$$2R \geq t(t^4 + 7t^2 + 9)(A_1 + A_2)$$

implies $\Theta_{PP}(t) \leq 4/[t(t^4 + 7t^2 + 9)]$. Since

$$\int_0^\infty \frac{4}{t^4 + 7t^2 + 9} dt = \frac{2\pi}{3\sqrt{13}},$$

the signed energy difference exceeds $-8/(3\sqrt{13})$, proving (13). This scalar theorem requires the actual common pivot and is not applied to multi-pivot shared-cut packets. $\square$

Join cyclic blocks when they share a cut and call the components *shared-cut clusters*. Contract each cluster in the block-cut tree and retain the actual bridge paths between clusters; the resulting reduced cluster graph is a tree.

Lemma 2.3 (Whole-cluster realization). *Let $S_1, \dots, S_r$ be pairwise disjoint connected subtrees of the reduced cluster tree and suppose every marked shared-cut cluster belongs to exactly one S_i . Then $V(G)$ has a partition into connected induced territories $G_1, \dots, G_r$ such that G_i retains, whole, exactly the cyclic blocks in clusters assigned to S_i . Every separation is at an actual bridge, and every connector remnant, unmarked Steiner branch, connector entry, and off-hull tree has exactly one owner.*

Proof. Contract each S_i to a labelled root and assign every remaining reduced-tree vertex to a nearest root, breaking ties by a fixed label order. Each label class in a tree is connected. An edge between classes expands in the block-cut tree to a path of bridge blocks; cut one actual bridge on that path and give the two remnants to the adjacent owners. At a Steiner branch, give the branch vertex and one initial segment to one owner and cut an actual bridge on every other incident segment. A component off the cyclic hull has a unique hull attachment, or the block-cut graph would contain a cycle, and is assigned whole to that attachment's owner. The resulting territories are connected, induced, disjoint, exhaustive, and retain the claimed whole clusters. $\square$

Lemma 2.4 (Accepted one-cycle interval realization). *Let one cycle C in a shared-cut cluster be selected by a finite certificate. Group its incidence branches and external connector entries by their vertex on C ; coincident marks form one group. If the distinct groups can be assigned to nonempty proper consecutive intervals of C , then each interval together with all branches and entries assigned*

to it is a connected induced territory, the territories partition everything assigned at C , and C is retained by none. A later accepted split may refine one territory. Every coincident entry at a shared cut follows the unique interval owning that cut.

Proof. List the distinct marked vertices cyclically. Cut one cycle edge in each chosen gap and assign every unmarked gap vertex to one adjacent interval. Each interval is a proper path. The cycle-cut incidence graph is a tree, so every branch of $I - C$ meets C at exactly its marked vertex and follows exactly one interval. Entries coincident with that vertex follow the same owner and do not demand an empty extra interval. A refinement partitions only one already induced vertex set; hence inducedness, disjointness, exhaustiveness, and unique ownership persist sequentially. $\square$

3 Exact DNN reduction

For a connected edge-containing graph set

$$\kappa(G) = \sup_{\substack{M \succeq 0, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [4, 3] says that for bridge-block count b and cycle lengths $\ell_1, \dots, \ell_8$,

$$\kappa(G) = b + \sum_i \kappa(C_{\ell_i}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}, \end{cases} \quad s^-(G) \leq \kappa(G).$$

Put $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ . Since an octacyclic connected cactus has $|E| = n + 7$, block counting gives

$$\sigma(G) \geq 7 - \sum_{i=1}^8 \epsilon_{\ell_i}. \quad (14)$$

The odd sequence decreases, $\epsilon_3 = 1$, and with $a = \epsilon_5 = 5 - 2\sqrt{5}$ the exact comparisons are

$$3a < 2, \quad 2a > 1, \quad \epsilon_5 + \epsilon_7 < 1.$$

Hence at most five triangles gives $\sum \epsilon_i \leq 5 + 3a < 7$. With exactly six triangles the remaining pair reaches one only for P, P . With at least seven triangles the multiset is $T^7 Q$. Therefore (14) is strict outside exactly

$$T^7 Q = \{3, 3, 3, 3, 3, 3, 3, q\}, \quad T^6 PP = \{3, 3, 3, 3, 3, 3, 5, 5\}. \quad (15)$$

4 Disconnected shared-cut graph

The census uses exact rational arithmetic from Python's `fractions` module and gives

residual	all partitions	proper partitions	direct packet rows
$T^7 Q$	45	44	42
$T^6 PP$	77	76	70

A direct row has a positive packet sum, with strictness tracked symbolically. The only structural rows are

$$Q|T|T|T|T|T|T, \quad Q|T^7, \quad (16)$$

$$\begin{aligned} P|P|T|T|T|T|T|T, \quad P|T|T|T|T|T^2P, \quad P|T|T|T|T^3P, \\ P|T|T|T^4P, \quad P|T|T^5P, \quad P|T^6P. \end{aligned} \quad (17)$$

4.1 The two T^7Q rows

In the all-singleton row, a singleton triangle is a leaf of the finite reduced tree; cutting its first bridge gives a strict triangle and a connected heptacyclic complement. Consider $T^7|Q$. If the triangle incidence tree is not a bouquet, split an internal triangle at its cyclic marks and, when needed, at the private connector-entry mark. If the Q branch retains r triangles, $r \geq 3$ is positive by lower rank, $r = 2$ is nonnegative with another strict triangle branch, and $r = 1$ is a positive TQ packet. If $r = 0$, the entry uses one of at most three marks, so six triangles occupy at most two other branches; one contains at least three triangles and has surplus > 2 , absorbing the isolated hostile loss > -1 .

If the seven triangles form a bouquet but the connector enters a private triangle vertex, splitting there gives $A_6 + Q$ and surplus $> 1 - \delta_q > 0$. The only locked structure is

$$\text{seven triangles sharing } x, \text{ joined from } x \text{ to } Q \text{ by an arbitrary positive connector.} \quad (18)$$

Its triangles are pairwise intersecting, so their vertex-packing number is exactly one. Lemma 2.1, with $a = 7$, applies directly and gives

$$\sigma(G) > 7 - \delta_q > 0.$$

This verifies the standalone lemma's hypothesis on the packet itself; it is not the retracted all-rank guard. Nonhostile Q also follows from A_7 strictness and a bridge partition. Thus all $44 = 42 + 2$ proper T^7Q rows are proved.

4.2 The T^6PP rows before the locked endpoint

For the first five rows of (17), a singleton-triangle leaf gives T and a heptacyclic complement. If none is a leaf, the two nontriangle endpoint clusters are the leaves of their path. Pair the singleton triangle nearest the singleton pentagon with that pentagon, obtaining a positive TP terminal packet and a strict hexacyclic complement. The separate row $P_0|T^6|P_1$ has

$$\sigma(G) > \sigma(A_6) - 2\delta > 1 - 2\delta = 5 - 2\sqrt{5} > 0.$$

Only $T^6P_0|P_1$ remains. Splitting an internal P_0 into proper intervals gives a mixed lower-rank entry branch and another strict triangular branch. If P_0 is an incidence leaf but the connector enters a private point of P_0 , split between entry and cyclic cut to obtain $A_6 + P_1 > 1 - \delta$. The remaining entry-locked family is denoted $G6PP$.

4.3 Strict-last-bridge $G6PP$: $877 = 861 + 16$

Cut the last actual bridge before remote P_1 . Its whole side is pentagonal unicyclic and contributes at least $-\delta$. The entire connector remnant on the T^6P_0 side stays there as a tree rooted at its marked first cyclic-hull entry. Nothing is moved across the cut. The color-preserving census has

$$226 \text{ unrooted } T^6P_0 \text{ incidences, } 111 \text{ with } P_0 \text{ a leaf, } 877 \text{ marked root classes.}$$

This is the exact root universe. It starts with all 226 color-preserving bipartite cycle-cut trees on colors T^6P_0 , restricts to the 111 trees in which P_0 has incidence degree one, and marks the first cyclic-hull entry modulo color-preserving automorphism. A mark is either any shared cut, including the unique cut of P_0 , or a private triangle vertex. A triangle of incidence degree d contributes $3 - d$ labelled private positions before orbit quotienting. Private vertices of P_0 are absent because the entry-locked reduction has already disposed of that case. Thus every possible locked connector entry, and no unlocked entry, occurs among the 877 classes. The conservative one-router test proves 861 and leaves exactly 16, by cut number $c = 1, \dots, 6$:

$$(2, 5, 5, 4, 0, 0).$$

Here are the exact sixteen marked profiles. Labels $0, \dots, 5$ are triangles, $6 = P_0$; R marks a cut root and TR a private triangle root.

class	c	canonical rooted code
L1	1	R(P()T()T()T()T()T())
L2	1	X(P()T()T()T()T()TR())
L3	2	T(R(P()T()T()T()T())X(T()))
L4	2	T(R(T()))X(P()T()T()T()T())
L5	2	T(X(P()T()T()T()T())X(TR()))
L6	2	T(X(P()T()T()T()TR())X(T()))
L7	2	TR(X(P()T()T()T()T())X(T()))
L8	3	R(P()T()T()T(X(T)))T(X(T()))
L9	3	X(P()T()T()T(R(T)))T(X(T()))
L10	3	X(P()T()T()T(X(T)))T(X(TR()))
L11	3	X(P()T()T()T(X(T)))TR(X(T()))
L12	3	X(P()T()T(X(T)))T(X(T()))TR()
L13	4	R(P()T(X(T)))T(X(T))T(X(T)))
L14	4	X(P()T(R(T)))T(X(T))T(X(T)))
L15	4	X(P()T(X(T)))T(X(T))T(X(TR()))
L16	4	X(P()T(X(T)))T(X(T))TR(X(T()))

The independent replacement verifier proves the following exhaustive ledgers. E is an acyclic private-entry interval, charged conservatively by $\sigma(E) \geq -1$.

classes	clustered-side packets	total with P_1
L1–L2	common-cut T^6P_0	$> 6 - 2\delta$
L3–L6	$T +$ common-cut T^4P_0	$> 4 - 2\delta$
L7	$T +$ common-cut $T^4P_0 + E$	$> 3 - 2\delta$
L8–L10, L12	$T + T +$ common-cut T^2P_0	$> 2 - 2\delta$
L11	$T + T +$ common-cut $T^2P_0 + E$	$> 1 - 2\delta$
L13–L16	$T + T +$ shared-cut TTP_0	$> 2 - 2\delta$

For L13–L16 the two retained triangles share an actual cut, but P_0 need not contain it; this is precisely the shared-cut TTP distinction. Its bound is $> 2 - \delta$, not an invocation of the common-pivot theorem. The weakest case is L11:

$$0 + 0 + (2 - \delta) - \delta - 1 = 1 - 2\delta = 5 - 2\sqrt{5} > 0.$$

Binary routers are divided into a marked singleton and complementary edge; saturated routers have three forced singleton intervals. Every retained cut, entry remnant, and attached tree has one owner. More explicitly, L3–L7 split T_0 and retain T_2 plus $T_1T_3T_4T_5P_0$; L8–L12 split T_0 first and then T_1 inside the hub-owning territory, retaining $T_2, T_5, T_3T_4P_0$; L13 splits the same routers and retains $T_2, T_4, T_3T_5P_0$; L14–L16 split T_1 and then T_3 , retaining $T_4, T_5, T_0T_2P_0$. At each first split the branches of the deleted router are disjoint components; the second split is performed only inside the unique component owning the retained hub. Every retained cut is incident only with cycles in one final packet, while a private entry on a deleted router becomes the separate acyclic E owner exactly in L7 and L11. Lemma 2.4 therefore proves sequential ownership, not merely simultaneous ownership in the incidence code. Thus all $877 = 861 + 16$ classes close, and all $76 = 70 + 6$ disconnected T^6PP rows are proved.

5 One fully shared T^7Q cluster

The exact color-preserving incidence census, by cut number $c = 1, \dots, 7$, is

Q capacity	1	2	3	4	5	6	7	total
$q = 3$	1	9	49	142	236	191	60	688
$q = 4$	1	9	49	145	243	202	66	715
$q = 5$	1	9	49	145	245	205	69	723
$q = 6$	1	9	49	145	245	206	70	725
$q \geq 7$	1	9	49	145	245	206	71	726

For every nonbouquet type, the exact ordinary-split ledger finds a legal cycle split into proper consecutive intervals and a positive packet sum. Its branches use only A_r , isolated Q , TQ , TTQ , and rank four through seven packets, with all retained shared-cut hypotheses checked. The unique exception in every capacity regime is the one-cut bouquet in which all seven triangles and Q share one vertex. Lemma 2.2 gives there

$$\sigma(T^7Q) > 7 - \delta_q > 6 \quad \text{for hostile } Q, \quad \sigma(T^7Q) > 7 \quad \text{otherwise.}$$

The theorem is used only in its actual common-pivot scope. Hence every fully shared T^7Q cactus is proved.

6 One fully shared T^6PP cluster: $2116 = 2110 + 6$

The exact color-preserving census is

	$c = 1$	$c = 2$	$c = 3$	$c = 4$	$c = 5$	$c = 6$	$c = 7$	total
all	1	14	106	377	728	657	233	2116
ordinary-split safe	0	13	104	376	727	657	233	2110
exceptions	1	1	2	1	1	0	0	6

Each of the 2110 accepted types has an exact positive packet ledger and a legal proper-interval split. The six remaining color-preserving types admit the following complete replacements:

code	replacement packetization	certified surplus
U1	common-cut T^6PP	$> 7 - 4/(3\sqrt{13})$
U2	$P +$ common-cut T^5P	$> 5 - 2\delta$
U3	$P + T +$ common-cut T^4P	$> 4 - 2\delta$
U4	$P + P + A_4$	$> 3 - 2\delta$
U5	$P + P + T + A_3$	$> 2 - 2\delta$
U6	$P + P + T + T + A_2$	$> 1 - 2\delta$

For completeness, the router realizations are as follows. U1 is exactly the common-cut (6,2) bouquet and uses (13) without splitting its pivot. In U2, a binary triangle router separates one pentagon and leaves an actual common-cut T^5P . In U3, one saturated router separates P , T , and an actual common-cut T^4P . In U4, two binary routers isolate the two pentagons and leave A_4 . In U5, a saturated router and then a binary router leave $P + P + T + A_3$. In U6, two successive saturated-router splits leave $P + P + T + T + A_2$.

At a binary router, the marked pentagon-side vertex is a singleton and the other two router vertices form the complementary interval. At a saturated router all three marks occupy distinct triangle vertices, so the singleton intervals are forced. Each pentagon's unique cut remains with it; the common hub has one owner; a second split only refines that owner's territory. Thus these descriptions cover every cyclic order and every assignment of attached trees. The weakest resolution is exactly

$$\sigma(G) > -\delta - \delta + 0 + 0 + 1 = 1 - 2\delta = 5 - 2\sqrt{5} > 0.$$

Therefore all $2116 = 2110 + 6$ fully shared T^6PP types are proved without any rooted hostile-cycle guard.

7 Main theorem and exhaustion

Theorem 7.1. *Every connected octacyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The DNN inequality (14) proves every cycle multiset except the two in (15). If the shared-cut graph is disconnected, the exact $44 = 42 + 2$ and $76 = 70 + 6$ partition audit, the directly verified packing-one $G7Q$ endpoint, and the strict $877 = 861 + 16$ last-bridge certificate prove both residual families. If the shared-cut graph is connected, all eight cycles form one incidence-tree cluster. The fully shared T^7Q census and common-cut theorem prove its every incidence; the $2116 = 2110 + 6$ census and the six router resolutions prove every T^6PP incidence.

Every reduced-tree separation is at an actual bridge. In $G6PP$, the remote P_1 cut leaves all other connector vertices on the marked cluster side. In $G7Q$, Lemma 2.1 itself permits arbitrary positive path length and trees on internal path vertices. Every cycle split uses nonempty proper consecutive intervals; every shared cut, incidence branch, connector remnant, and off-hull tree has exactly one owner. Therefore all territories are induced, disjoint, and exhaustive, and all packet hypotheses persist. No edge monotonicity, tree relocation, retracted all-rank guard, or superseded uncut $868 + 9$ certificate occurs. In every case $\sigma(G) > 0$. $\square$

Since $|E(G)| = n + 7$, this proves the strict conclusion of [2, Conjecture 1.2] for every connected octacyclic cactus.

A Exact reproduction and dependencies

The census and router certificates use the Python standard library and exact integer or rational arithmetic from Python’s `fractions` module. The final symbolic coefficient certificate requires SymPy 1.14.0. An isolated setup from the repository root is

```
python3 -m venv /tmp/opencode/octacyclic-cacti-venv
/tmp/opencode/octacyclic-cacti-venv/bin/python -m pip install \
  --upgrade pip
/tmp/opencode/octacyclic-cacti-venv/bin/python -m pip install \
  sympy==1.14.0
```

Then run, in this order,

```
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-disconnected-partition-census.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-t6p-last-bridge-conservative.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-g6pp-last-bridge-census.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-t6p-last-bridge-sixteen-resolution.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-g6pp-last-bridge-four-resolution.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-fully-shared-incidence-census.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  research/octacyclic-t6pp-six-exceptions-resolution.py
/tmp/opencode/octacyclic-cacti-venv/bin/python \
  positive-square-energy/experiments/c5_bouquet_matching_certificate.py
```

The first command asserts 45, 44, 42, 2 and 77, 76, 70, 6. The second and fourth regenerate $877 = 861 + 16$, the profile distribution $(2, 5, 5, 4, 0, 0)$, all L1–L16 codes, 16/16 closure, and weakest margin $1 - 2\delta$. The third and fifth are independent cross-checks. The sixth enumerates the fully shared tables and applies the encoded one-cycle SAFE test; it reports the bouquet and U1–U6 as the unresolved finite incidence types. The new seventh verifier regenerates those six rows, checks their stated edge sets, router interval sizes, sequential branch refinements, retained-cycle packet sets, common-cut hypotheses, cyclic-cut ownership, and exact symbolic ledgers, and reports 6/6 closure. It imports the census module, so it is independent of neither the incidence generator nor the SAFE classifier; it does not verify the analytic packet inequalities. Likewise, the conservative last-bridge script imports the marked-census module for finite incidence generation, root orbits, and encoded packet bounds; the separate sixteen-resolution and independent G6PP scripts supply ownership and census cross-checks. None of these scripts proves induced realization for arbitrary connectors or attached trees: Lemmas 2.3 and 2.4 do that. The last command checks only the 1290-term positive-coefficient inequality used in the two-pentagon common-cut theorem.

Compile from the repository root with

```
bash scripts/build-paper.sh all-octacyclic-cacti
```

or, after installing the packages listed in the accompanying README, with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -cd all-octacyclic-cacti/paper.tex
```

References

- [1] S. Akbari, H. Kumar, B. Mohar, and S. Pragada, *A linear lower bound for the square energy of graphs*, Electron. J. Combin. 32(3) (2025), Paper No. P3.53.
- [2] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264, 2025.
- [3] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.
- [4] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; repository path `sharp-cactus-dnn/paper.tex`.
- [5] Companion manuscript, *Positive Square Energy of Every Bicyclic Cactus*, 2026; repository path `all-bicyclic-cacti/paper.tex`.
- [6] Companion manuscript, *Positive Square Energy of Every Tricyclic Cactus*, 2026; repository path `all-tricyclic-cacti/paper.tex`.
- [7] Companion manuscript, *Positive Square Energy of Every Tetracyclic Cactus*, repaired strict theorem, 2026; repository path `all-tetracyclic-cacti/paper.tex`.
- [8] Companion manuscript, *Positive Square Energy of Every Pentacyclic Cactus*, 2026; repository path `all-pentacyclic-cacti/paper.tex`.
- [9] Companion manuscript, *Positive Square Energy of Every Hexacyclic Cactus*, 2026; repository path `all-hexacyclic-cacti/paper.tex`.
- [10] Companion manuscript, *Positive Square Energy of Every Heptacyclic Cactus*, 2026; repository path `all-heptacyclic-cacti/paper.tex`.
- [11] Companion manuscript, *Square Energy from Cycle Packing and Block-Graph Territories*, 2026; repository path `packing-two-square-energy/paper.tex`.
- [12] *Positive square energy of every connected octacyclic cactus*, corrected synthesis, 26 July 2026; repository path `research/octacyclic-cactus-complete-synthesis-2026-07-26.md`.
- [13] *Rooted packing-one hostile-cycle lemma for octacyclic provenance*, 26 July 2026; repository path `research/octacyclic-packing-one-hostile-cycle-lemma-2026-07-26.md`.
- [14] *Conservative last-bridge rerun: all sixteen failed classes*, 26 July 2026; repository path `research/octacyclic-t6p-last-bridge-conservative-resolution-2026-07-26.md`.
- [15] *Exact fully shared octacyclic incidence census: T^7Q and T^6PP* , 26 July 2026; repository path `research/octacyclic-fully-shared-incidence-census-2026-07-26.md`.
- [16] *The six fully shared octacyclic T^6PP census exceptions: replacement audit*, 26 July 2026; repository path `research/octacyclic-t6pp-six-exceptions-resolution-2026-07-26.md`.
- [17] *Common-cut residual bouquets: an exact rooted Schur–Sachs inequality*, 26 July 2026; repository path `research/common-cut-bouquet-rooted-schur-2026-07-26.md`.